



NEXUS 2026

FORMULA HARDWARE

Organized by :-ETSA , PCCoE

Event Type :- Technical - Hardware Hackathon

Venue :- Third Floor Labs – E&TC Dept., PCCOE

Time :- 8:00 AM – 8:00 PM

Coordinators :- Samrat Latane , Vedant Jadhav , Yugandhar Suryawanshi , Parth Fundkar

Team Size :- 2–5 Members

Registration Fee :- ₹599 per team

Participation Details:-

- All students of PCCoE, across years and departments
- **Free for PCCOE Students**
- Participants may bring any project idea or problem statement of their choice.
- No compulsory common hardware theme will be imposed.
- **Registration fees:** ₹599 per team

Prize Pool:-

Position	Prize Amount
First Place	₹12,000
Second Place	₹8,000
Third Place	₹5,000
Total Prize Pool	₹25,000

Description:-

FORMULA HARDWARE is an open-theme hardware hackathon designed to encourage creativity, practical engineering, innovation and working prototypes. Teams may select any project idea, real-world problem statement, application area or technical domain, subject to safety and event guidelines.

The event is inspired by the **precision, pressure and teamwork of Formula 1**, while focusing on real hardware, technical evaluation and practical implementation.

01 EVENT OVERVIEW

- Participants may bring any project idea or problem statement of their choice.
- Projects may involve **electronics, embedded systems, programming, mechanics, AI/ML, IoT, automation, robotics, communication, sustainability or social innovation**.
- No compulsory common hardware theme will be imposed.
- The event has no racetrack, lap race, car-building requirement or compulsory racing concept.
- Teams will be evaluated on **technical clarity, feasibility, implementation, demonstration, teamwork and presentation**.

02 EVENT FORMAT

ROUND 1	ROUND 2
6-slide PPT Technical clarity & feasibility	12- Hour Prototype Build Build • Test • Integrate • Demonstrate

03 ROUND 1 — IDEA PRESENTATION

Shortlisted teams will first present their proposed project idea to the evaluation panel.

- Submit a 6-slide PowerPoint presentation.
- The PPT must clearly explain the problem statement, proposed solution, working principle and expected outcome.
- Teams should mention the major hardware, software, sensors, controllers, actuators or other resources planned.
- Shortlisting will be based on technical clarity and feasibility.
- Only shortlisted teams will proceed to Round 2.
- Teams should be prepared to answer questions from judges regarding practicality, implementation and limitations.

Suggested PPT structure

- Slide 1: Problem statement and motivation
- Slide 2: Proposed solution and innovation
- Slide 3: System architecture / working principle
- Slide 4: Components and technical requirements
- Slide 5: Feasibility, implementation plan and testing
- Slide 6: Expected outcome, impact and future scope

04 ROUND 2 — 12-HOUR PROTOTYPE BUILD & EVALUATION

Shortlisted teams will develop, assemble, code, test and demonstrate a working hardware prototype during the 12-hour event window.

- **Event duration:** 12 hours, from 8:00 AM to 8:00 PM.
- Teams must bring their **own components, tools, wires, consumables, batteries** and **project-specific materials**.
- Teams may build a new project or improve an existing concept, but prior work must be disclosed when requested.
- Judges may ask for **architecture explanation, live demonstration, subsystem testing and technical justification**.
- Partially functional projects may demonstrate working subsystems with an honest explanation of limitations.

EVENT FLOW : Round 2

Time	Activity	Description
8:00 AM – 8:30 AM	Reporting & Registration	Participant reporting, ID verification, team confirmation, and lab/classroom allocation.
8:30 AM – 9:45 AM	Prototype Build – Phase 1	Initial setup including component understanding, circuit planning, and basic assembly.
10:00 AM – 10:15 AM	Breakfast Break	Breakfast for all participants.
10:15 AM – 12:00 PM	Prototype Build – Phase 2	Circuit refinement, initial coding, and implementation of core functionality.
12:00 PM – 12:30 PM	Evaluation Round 1 – Idea & Design Review	Review of problem statement, system design, feasibility, component selection, and execution plan.
12:30 PM – 1:30 PM	Lunch Break	Scheduled lunch interval.
1:30 PM – 5:00 PM	Prototype Build – Phase 3	Advanced development, debugging, testing, and incorporation of evaluator feedback.
5:00 PM – 5:15 PM	High Tea Break	Short refreshment break.
5:15 PM – 7:00 PM	Final Build Phase	Final integration, optimization, and demo preparation.
7:00 PM – 8:00 PM	Evaluation Round 2 – Final Demonstration	Live prototype demonstration and final judging based on functionality and execution.

Special Challenges

- **SURPRISE CONSTRAINT TWIST**

Teams may be given an unexpected technical or resource constraint during the building. They must adapt their design while maintaining safety and engineering logic.

- **REVERSE ENGINEERING CHALLENGE**

Teams may be asked to inspect, interpret or explain an assigned circuit, module, mechanism, signal path or system behavior.

- **BONUS MARKS**

Additional marks may be awarded for clearly explaining system logic, signal flow, fault detection, failure handling, debugging approach and design trade-offs.

Round 2 evaluation focus:

- Working performance and live demonstration
- Technical architecture and integration
- Implementation quality, neatness and reliability
- Testing, debugging and validation
- Response to surprise constraints
- Reverse engineering understanding
- Teamwork, communication and technical presentation

05 ELIGIBILITY & TEAM FORMATION

- The event is open to engineering students.
- Each team must consist of **2 to 5 members**.
- Each member must have a clearly defined responsibility.
- Each team must nominate **one Team Captain** for official communication.
- All members are expected to contribute meaningfully and understand the project.

06 COMPONENTS, TOOLS & RESOURCES

- Every team is responsible for bringing its own components, modules, sensors, controllers, motors, batteries, mechanical parts, tools, wires, consumables and other project requirements.
- Organizers will **not** provide project-specific components unless announced before the event.
- Teams must verify component compatibility, ratings, battery condition and safe operation.
- Use of pre-built modules is permitted as part of a larger original system.
- Presenting a ready-made complete project as original work is prohibited.
- Any external assistance, online resources or AI tool used must be disclosed when requested.

07 SAFETY & ETHICAL ENGINEERING

- Follow **safe soldering, wiring, battery handling, charging** and **power-use practices**.
- All electrical connections must be insulated, secure and suitable for operating current.
- Battery terminals must be protected against short circuits, overheating and physical damage.
- Projects must not contain sharp, exposed, loose, explosive or intentionally hazardous parts.
- Organizers may stop or remove any unsafe project.
- Plagiarism, false claims, fabricated results or misrepresentation may lead to disqualification.

08 EVALUATION & JUDGING

Criterion	Evaluation focus
Problem Understanding	Clarity, relevance and definition of the selected problem
Innovation	Originality, creativity and quality of the approach
Technical Architecture	Integration of hardware, software, mechanics, AI/ML or related systems
Implementation Quality	Build quality, coding, integration, reliability and neatness
Functionality & Demonstration	Working performance, proof of concept and live demonstration
Testing & Validation	Testing method, observations, debugging and evidence
Presentation & Team Understanding	Clarity, documentation, teamwork and responses
Impact & Future Scope	Practical usefulness, scalability, sustainability and improvements

10 GENERAL RULES

- Only registered team members may represent a team during evaluation.
- Teams must follow venue rules and directions from organizers, judges, mentors and volunteers.
- Unauthorized interference with another team's project or workspace is prohibited.
- Fabricating test results or project claims is prohibited.
- The committee may make minor operational changes for safety and smooth conduct.
- The decision of the judges and organizing committee shall be final.

11 // EVENT DETAILS & CONTACT

Venue: E&TC Dept., PCCOE , Pune

COORDINATORS

Samrat Latane — 9146649309

Vedant Jadhav — 8308108646

Yugandhar Suryawanshi — 7875251257

Parth Fundkar — 7058355379

INNOVATE FREELY. BUILD RESPONSIBLY. DEMONSTRATE WITH IMPACT.

This event celebrates independent ideas, practical engineering, teamwork and the courage to build solutions to problems that matter.